

Shore

Student Number:

Set:

Year 12
Mathematics - Extension 2
Trial Examination
2009

General Instructions

- * Reading time – 5 minutes
- * Working time – 3 hours
- * Write using black or blue pen
- * Board-approved calculators may be used
- * All necessary working should be shown in every question
- * A table of standard integrals is attached on the final page

Note: Any time you have remaining should be spent revising your answers.

Total marks - 120

- * Attempt Questions 1 - 8
- * All questions are of equal value
- * Start each question in a new writing booklet
- * Write your examination number on the front cover of each booklet to be handed in
- * If you do not attempt a question, submit a blank booklet with your examination number and “N/A” on the front cover

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

Total marks - 120
Attempt Questions 1 - 8

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks) Use a SEPARATE writing booklet **Marks**

(a) Find the indefinite integrals:

(i) $\int \sec^4 x \, dx$ **2**

(ii) $\int \sqrt{1-x^2} \, dx$ **4**

(b) Consider the definite integral $I_n = \int_0^2 \frac{x^n}{x^3+1} \, dx$.

(i) Show that $I_2 = \frac{2}{3} \log_e 3$. **2**

(ii) Using your knowledge of factorisation and without evaluating more than one integral, show that

$$I_2 - I_1 + I_0 = \log_e 3$$
2

(iii) Using a similar approach to that used in (ii), show that

$$I_1 + I_0 = \frac{\pi}{\sqrt{3}}.$$
3

(iv) Using the above results or otherwise find the exact value of I_0 . **2**

Question 2 (15 marks) Use a SEPARATE writing booklet**Marks**

- (a) Make neat sketches of the following, showing all intercepts and asymptotes. There is no need to use calculus.

(i) $y = x^2(x-2)(x-3)$ 2

(ii) $y = \frac{1}{x^2(x-2)(x-3)}$ 2

(iii) $y = \frac{x^2}{(x-2)(x-3)}$ 2

(iv) $y = x\sqrt{(x-2)(x-3)}$ 2

(v) $y = x^2|x-2|(x-3)$ 2

- (b) Consider the equation $e^{2x} = k\sqrt{x}$.

(i) Explain why this equation has no solutions when $k \leq 0$. 1

(ii) Find the value of k for which the equation has exactly one real solution. 4

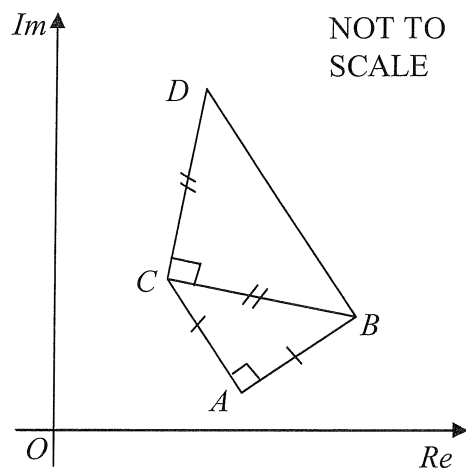
Question 3 (15 marks) Use a SEPARATE writing booklet

Marks

- (a) Given $z = 1 - 2i$ is a complex root of the quadratic equation $z^2 + (1 + i)z + k = 0$, find the other root and the value of k . **3**
- (b) Find all complex numbers $z = a + bi$, where a and b are real such that $|z|^2 - iz = 16 - 2i$. **3**
- (c) Consider all complex numbers z such that $\arg\left(\frac{z-1}{z-i}\right) = \frac{\pi}{4}$
- (i) Make a neat sketch of the locus of z showing important features. **2**
- (ii) Determine the exact maximum value of $|z|$. **1**
- (iii) Determine (in radians correct to 3 significant figures) the maximum value of $\arg(z+1)$. **3**

Question 3 continues on page 5

(d)



In the diagram, the points A , B , C and D represent the complex numbers z_1 , z_2 , z_3 and z_4 respectively. Both $\triangle ABC$ and $\triangle BCD$ are right angled isosceles triangles as shown.

- (i) Show that the complex number z_3 can be written as

$$z_3 = (1 - i)z_1 + iz_2.$$

1

- (ii) Hence express the complex number z_4 in terms of z_1 and z_2 , giving your answer in simplest form.

2

End of Question 3

Question 4 (15 marks) Use a SEPARATE writing booklet

Marks

- (a) Use Mathematical Induction to prove De Moivre's Theorem, ie

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta \quad \text{for all positive integers } n. \quad 4$$

- (b) The equation $x^3 + 3px - 1 = 0$, where p is real, has roots α , β and γ .

- (i) Show that the monic cubic equation, with coefficients in terms of p , whose roots are α^2 , β^2 and γ^2 is

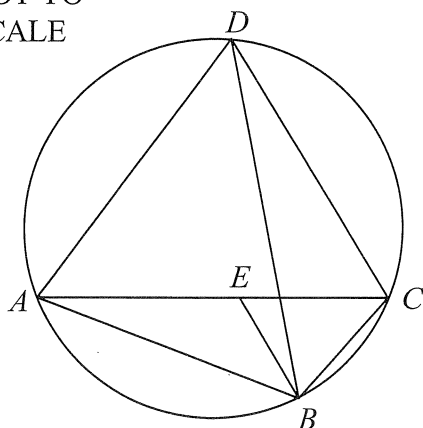
$$y^3 + 6py^2 + 9p^2y - 1 = 0. \quad 2$$

- (ii) Hence or otherwise obtain the monic cubic equation, with

coefficients in terms of p , whose roots are $\frac{\beta\gamma}{\alpha}$, $\frac{\gamma\alpha}{\beta}$ and $\frac{\alpha\beta}{\gamma}$. 3

- (c)

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The figure shows a cyclic quadrilateral $ABCD$ with diagonals AC and BD .

E is a point on AC such that $\angle ABE = \angle DBC$.

Make a neat copy of the diagram in your answer booklet.

- (i) Prove that $\triangle ABE \sim \triangle DBC$. 2

- (ii) Prove that $\triangle ABD \sim \triangle EBC$. 2

- (iii) Hence prove Ptolemy's Theorem, which is:

$$AB \cdot DC + AD \cdot BC = AC \cdot BD \quad 2$$

Question 5 (15 marks) Use a SEPARATE writing booklet**Marks**

- (a) Determine the eccentricity of the ellipse with equation

$$\frac{x^2}{16} + \frac{y^2}{25} = 1 \text{ and then make a neat sketch of the curve, clearly}$$

showing the coordinates of the foci and the equations of the directrices.

4

- (b) A point $P(a \sec \theta, b \tan \theta)$ lies on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

The line through P perpendicular to the x -axis meets an asymptote at Q and the normal at P meets the x -axis at N .

- (i) Make a neat sketch illustrating the information above.

1

- (ii) Show that the equation of the normal at P is

$$\frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2.$$

2

- (iii) Show that QN is perpendicular to the asymptote.

2

- (c) $P\left(p, \frac{1}{p}\right)$ and $Q\left(q, \frac{1}{q}\right)$ are two variable points on the rectangular hyperbola $xy = 1$ such that the chord PQ passes through the point $A(0, 2)$. M is the midpoint of PQ .

- (i) Show that PQ has equation $x + p q y - (p + q) = 0$ and hence deduce that $p + q = 2 p q$.

3

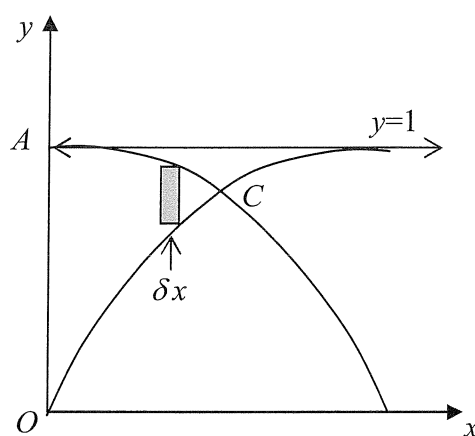
- (ii) You may assume that the tangent to $xy = 1$ at the point $(1, 1)$ passes through A . Determine the locus of M , being sure to state any restrictions on the domain.

3

- (a) The base of a solid is the region enclosed by the circle $x^2 + y^2 = 4$. Any cross sections of the solid formed by a plane perpendicular to the x -axis are equilateral triangles. Find the exact volume of the solid. **4**
- (b) (i) Make a neat sketch of the region enclosed between the curve $y = (x - 3)^2$ and the line $3x + y - 9 = 0$. Be sure to mark in the points of intersection. **2**
- (ii) The shaded region in (i) is rotated about the line $x = 3$. Use the method of cylindrical shells to find the exact volume of the solid generated. **3**

Question 6 continues on page 9

- (c) The diagram below shows part of the graphs of $y = \cos x$ and $y = \sin x$. The graph of $y = \cos x$ meets the y -axis at A , and C is the first point of intersection of the two graphs to the right of the y -axis. The region OAC is to be rotated about the line $y = 1$.



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- (i) Write down the coordinates of the point C . 1
- (ii) The shaded strip of width δx shown in the diagram is rotated about the line $y = 1$. Show that the volume δV of the resulting slice is given by
- $$\delta V = \pi(2 \cos x - 2 \sin x - \cos 2x) \delta x. \quad 2$$
- (iii) Hence find the exact volume of the solid formed when the region OAC is rotated about the line $y = 1$. 3

End of Question 6

Question 7 (15 marks) Use a SEPARATE writing booklet**Marks**

(a) (i) Prove that $\tan^{-1}(n+1) - \tan^{-1}(n) = \cot^{-1}(1+n+n^2)$ 2

(ii) Hence find the sum of the finite series

$$\cot^{-1}(3) + \cot^{-1}(7) + \cot^{-1}(13) + \dots + \cot^{-1}(1+n+n^2)$$

Give your answer in simplest form. 2

(b) You are given that for the complex number $z = \cos \theta + i \sin \theta$
and for positive integers n , the following results are true:

$$z^n + \frac{1}{z^n} = 2 \cos n\theta \quad \text{and} \quad z^n - \frac{1}{z^n} = 2i \sin n\theta$$

(i) Expand $\left(z + \frac{1}{z}\right)^4 + \left(z - \frac{1}{z}\right)^4$ and hence show that

$$4 \cos^4 \theta + 4 \sin^4 \theta = \cos 4\theta + 3 \quad \text{3}$$

(ii) By letting $x = \cos \theta$, show that the equation

$$8x^4 + 8(1-x^2)^2 = 7 \quad \text{has roots } x = \pm \cos \frac{\pi}{12}, \pm \cos \frac{5\pi}{12}. \quad \text{3}$$

(iii) Deduce that $\cos \frac{\pi}{12} \cos \frac{5\pi}{12} = \frac{1}{4}$ and $\cos \frac{\pi}{12} + \cos \frac{5\pi}{12} = \sqrt{\frac{3}{2}}$. 3

(iv) Hence or otherwise express $\cos \frac{\pi}{12}$ in surd form. 2

- (a) Six letters are chosen from the word AUSTRALIA. These six letters are then placed alongside one another to form a six letter arrangement. Find the number of distinct six letter arrangements which are possible, considering all choices.

4

- (b) It is given that for three positive real numbers a , b and c ,

$$\frac{a+b+c}{3} \geq \sqrt[3]{abc}$$

If we also know that $a+b+c=1$, prove that

(i) $\frac{1}{abc} \geq 27$

1

(ii) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9$

2

(iii) $(1-a)(1-b)(1-c) \geq 8abc$

2

(c) Let $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$.

(i) Show that $I_n = \left(\frac{n-1}{n} \right) I_{n-2}$ for $n \geq 2$.

3

(ii) Hence show that $\int_0^{\frac{\pi}{2}} \sin^{2n} x \, dx = \frac{\pi (2n)!}{2^{2n+1} (n!)^2}$

3

END OF EXAM

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0 \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

Note: $\ln x = \log_e x, x > 0$

$$\begin{aligned} \textcircled{1} (a) (i) I &= \int \sec^4 x \, dx \\ &= \int \sec^2 x (1 + \tan^2 x) \, dx \\ &= \underline{\underline{\tan x + \frac{1}{3} \tan^3 x + C}} \quad 2 \end{aligned}$$

$$\begin{aligned} (ii) I &= \int \sqrt{1-x^2} \, dx \quad \text{let } x = \sin \theta \\ &\quad dx = \cos \theta \, d\theta \\ &= \int \sqrt{1-\sin^2 \theta} \cos \theta \, d\theta \\ &= \int \cos^2 \theta \, d\theta \\ &= \frac{1}{2} \int (1 + \cos 2\theta) \, d\theta \\ &= \frac{1}{2} \theta + \frac{1}{4} \sin 2\theta + C \\ &= \underline{\underline{\frac{1}{2} \sin^{-1} x + \frac{1}{2} x \sqrt{1-x^2} + C}} \quad 4 \end{aligned}$$

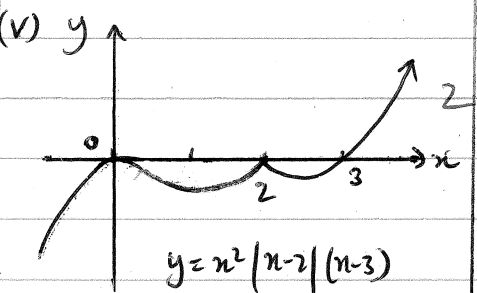
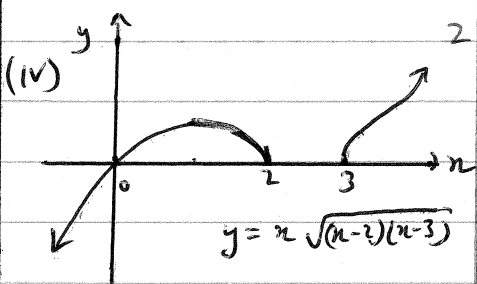
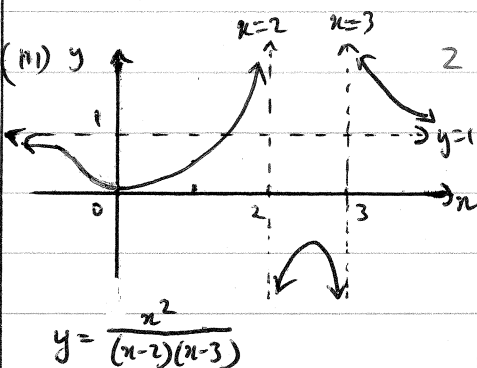
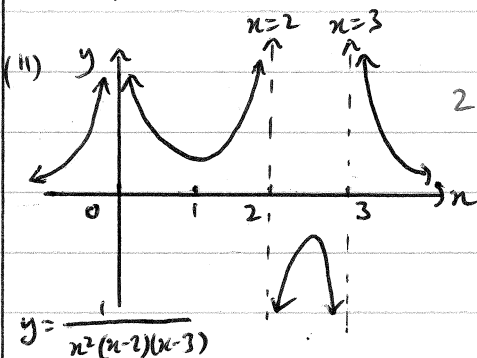
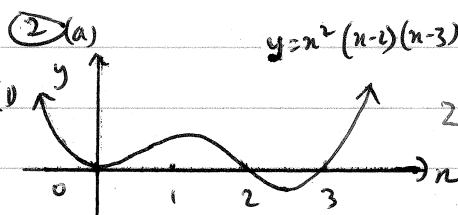
$$(b) I_n = \int_0^2 \frac{x^n}{x^3+1} \, dx$$

$$\begin{aligned} (i) I_2 &= \int_0^2 \frac{x^2}{x^3+1} \, dx \\ &= \left[\frac{1}{3} \log_e (x^3+1) \right]_0^2 \\ &= \frac{1}{3} (\log_e 9 - \log_e 1) \\ &= \underline{\underline{\frac{2}{3} \log_e 3}} \quad 2 \end{aligned}$$

$$\begin{aligned} (ii) I_2 - I_1 + I_0 &= \int_0^2 \frac{x^2 - x + 1}{x^3+1} \, dx \\ &= \int_0^2 \frac{1}{x+1} \, dx \\ &= [\log_e (x+1)]_0^2 \\ &= \underline{\underline{\log_e 3}} \quad 2 \end{aligned}$$

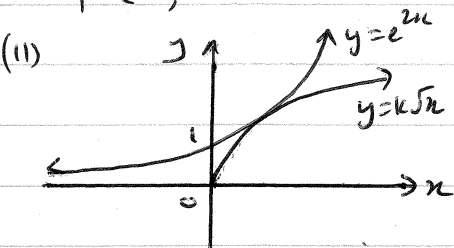
$$\begin{aligned} (iii) I_1 + I_0 &= \int_0^2 \frac{x+1}{x^3+1} \, dx \\ &= \int_0^2 \frac{1}{x^2-x+1} \, dx \\ &= \int_0^2 \frac{1}{\frac{5}{4} + (x-\frac{1}{2})^2} \, dx \\ &= \frac{2}{\sqrt{3}} \left[\tan^{-1} \left(\frac{2x-1}{\sqrt{3}} \right) \right]_0^2 \\ &= \frac{2}{\sqrt{3}} \left(\tan^{-1} \sqrt{3} - \tan^{-1} \left(-\frac{1}{\sqrt{3}} \right) \right) \\ &= \frac{2}{\sqrt{3}} \left(\frac{\pi}{3} + \frac{\pi}{6} \right) \\ &= \underline{\underline{\frac{\pi}{\sqrt{3}}}} \quad 3 \end{aligned}$$

$$\begin{aligned} (iv) I_0 &= \frac{1}{2} [I_1 + I_0 + I_2 - I_1 + I_0 - I_2] \\ &= \frac{1}{2} \left[\frac{\pi}{\sqrt{3}} + \log_e 3 - \frac{2}{3} \log_e 3 \right] \\ &= \underline{\underline{\frac{\pi}{2\sqrt{3}} + \frac{1}{6} \log_e 3}} \quad 2 \end{aligned}$$



$$(b) e^{2x} = k\sqrt{x}$$

(i) $e^{2x} > 0$ for all x , $\sqrt{x} \geq 0$
So if $k \leq 0$, $e^{2x} \neq k\sqrt{x}$



For one real solution $y = e^{2x}$ and $y = k\sqrt{x}$ have one point of intersection and share a common tangent.

$$\begin{aligned} y &= e^{2x}, & y &= k\sqrt{x} \\ y' &= 2e^{2x}, & y' &= \frac{k}{2\sqrt{x}} \end{aligned}$$

$$\Rightarrow 2k\sqrt{x} = \frac{k}{2\sqrt{x}}$$

$$\Rightarrow 4x = 1 \Rightarrow x = \frac{1}{4}$$

$$\Rightarrow e^{\frac{1}{2}} = k\sqrt{\frac{1}{4}}$$

$$\Rightarrow \underline{\underline{k = 2\sqrt{e}}} \quad 4$$

$$\textcircled{3} (a) z^2 + (1+i)z + k = 0$$

One root $1-2i$, let other be α

$$\alpha + 1-2i = -1-i \Rightarrow \underline{\underline{\alpha = -2+i}} \quad 1$$

$$\begin{aligned} \therefore k &= (1-2i)(-2+i) \\ &= -2+i+4i+2 \\ &= \underline{\underline{5i}} \quad 2 \end{aligned}$$

$$(b) |z|^2 - 1z = 16-2i, \quad z = a+bi$$

$$a^2+b^2 - ai + b = 16-2i$$

$$\Rightarrow a^2+b^2+b = 16 \text{ and } \underline{\underline{a=2}}$$

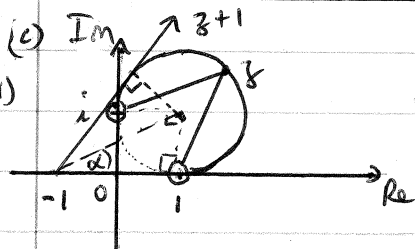
$$\therefore 2^2+b^2+b = 16$$

$$b^2+b-12=0$$

$$(b+4)(b-3)=0$$

$$\underline{\underline{b=-4 \text{ or } 3}}$$

$$\therefore \underline{\underline{z = 2+3i \text{ or } 2-4i}} \quad 3$$



(ii) $\text{Max } |z| = \sqrt{2} + 1$

(iii) $\text{Max } \arg(z+1) = 2 \tan^{-1} \frac{1}{2}$
 $= 0.868$

(d) (i) $\vec{AC} = i \cdot \vec{AB}$

$$z_3 - z_1 = i(z_2 - z_1)$$

$$z_3 = (1-i)z_1 + iz_2$$

(ii) $\vec{CD} = i \cdot \vec{CB}$

$$z_4 - z_3 = i(z_2 - z_3)$$

$$z_4 = iz_2 + (1-i)z_3$$

$$= iz_2 + (1-i)^2 z_1 + i(1-i)z_2$$

$$= -2iz_1 + (1+2i)z_2$$

(4) (R) RTP $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$

$n=1$ LHS = $\cos \theta + i \sin \theta$ = RHS $\therefore$ True

Assume $(\cos \theta + i \sin \theta)^k = \cos k\theta + i \sin k\theta$

Now $(\cos \theta + i \sin \theta)^{k+1} = (\cos \theta + i \sin \theta)(\cos k\theta + i \sin k\theta)$

$$= \cos \theta \cos k\theta - \sin \theta \sin k\theta + i(\cos \theta \sin k\theta + \sin \theta \cos k\theta)$$

$$= \cos(k+1)\theta + i \sin(k+1)\theta$$

$\therefore$ If true for $n=k$, also true for $n=k+1$

Since true when $n=1$, also when $n=1+1=2$ and so on

(b) (i) $x^3 + 3px - 1 = 0$ has roots α, β, γ

Let $y = x^2$, $x = \pm \sqrt{y}$

$$(\pm \sqrt{y})^3 + 3p(\pm \sqrt{y}) - 1 = 0$$

$$\pm \sqrt{y}(y + 3p) = 1$$

$$y(y^2 + 6py + 9p^2) = 1$$

$$y^3 + 6py^2 + 9p^2y - 1 = 0$$

has roots z, p^2, y

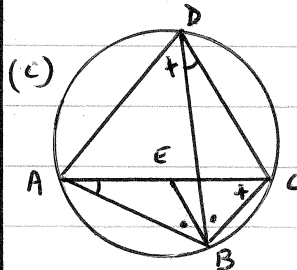
(ii) $\alpha\beta\gamma = 1 \Rightarrow \frac{\beta\gamma}{\alpha} = \frac{1}{\alpha^2}$ etc

$\therefore$ Let $z = \frac{1}{y} \Rightarrow y = \frac{1}{z}$

$$\left(\frac{1}{z}\right)^3 + 6p\left(\frac{1}{z}\right)^2 + 9p^2\left(\frac{1}{z}\right) - 1 = 0$$

$$1 + 6pz + 9p^2z^2 - z^3 = 0$$

$\therefore z^3 - 9p^2z^2 - 6pz - 1 = 0$ roots $\frac{\beta\gamma}{\alpha}$ etc.



Given
 $\angle ABE = \angle DBC$

(i) $\angle ABE = \angle DBC$ (Given)

$\angle BAE = \angle BDC$ (L's in same segment)

$\therefore \triangle ABE \parallel \triangle DBC$ (2 pairs of equal L's)

(ii) $\angle ABD = \angle ABE + \angle EBD$

$\angle ECB = \angle DBC + \angle EBD$

$\therefore \angle ABD = \angle ECB$

Also $\angle ADB = \angle ECB$ (L's in same segment)

$\therefore \triangle ABD \parallel \triangle ECB$ (2 pairs of equal L's)

(iii) $\frac{AB}{DB} = \frac{AE}{DC}$ (matching sides in $\triangle$ s)

$\Rightarrow AB \cdot DC = AE \cdot DB$ (1)

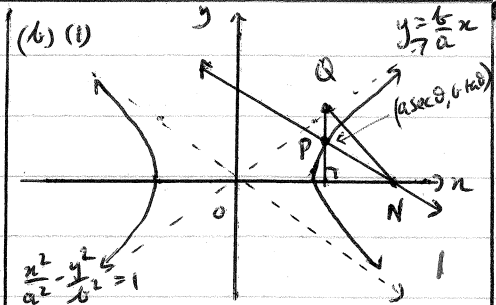
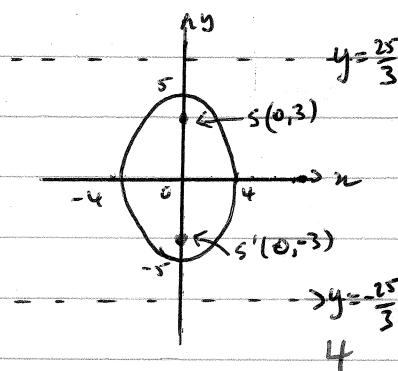
Also $\frac{AD}{EC} = \frac{BD}{BC}$ (matching sides in $\triangle$ s)

$\Rightarrow AD \cdot BC = EC \cdot BD$ (2)

(1) + (2) $AB \cdot DC + AD \cdot BC = (AE + EC) \cdot BD$

$= AC \cdot BD$

(5) (a) $\frac{x^2}{16} + \frac{y^2}{25} = 1$, $e = \sqrt{1 - \frac{16}{25}} = \frac{3}{5}$



(b) (i) At P, $x = a \sec \theta$, $dx = a \sec \theta \tan \theta d\theta$

$y = b \tan \theta$, $dy = b \sec^2 \theta d\theta$

$\frac{dy}{dx} = \frac{b \sec \theta}{a \tan \theta}$

Normal: $y - b \tan \theta = -\frac{a \tan \theta}{b \sec \theta} (x - a \sec \theta)$

$b y \sec \theta - b^2 \sec \theta \tan \theta = -a \tan \theta + a^2 \sec \theta \tan \theta$

$\therefore \frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2$

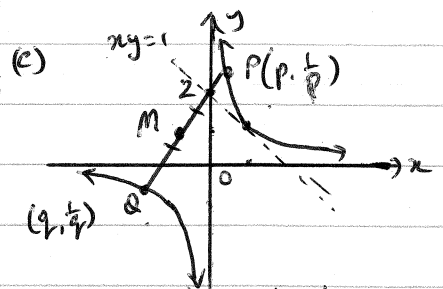
(iii) At Q, $x = a \sec \theta$, $y = b \sec \theta$

At N, $y = 0 \Rightarrow x = \frac{\sec \theta}{a} (a^2 + b^2)$

Gradient QN = $\frac{0 - b \sec \theta}{a \sec \theta + \frac{b^2 \sec \theta}{a} - a \sec \theta}$

$= -\frac{a}{b}$

Asymptote has gradient $\frac{b}{a} \Rightarrow \perp$ to QN



(i) PQ: $\frac{y - \frac{1}{p}}{x - p} = \frac{\frac{1}{q} - \frac{1}{p}}{q - p}$ (x, y)

$(py - q)(q - p) = (x - p)(p - q)$

$\Rightarrow x + pqy - (p+q) = 0$

subst (0,2) $\Rightarrow 2pq = p+q$

(ii) M is $(\frac{1}{2}(p+q), \frac{1}{2}(\frac{1}{p} + \frac{1}{q}))$

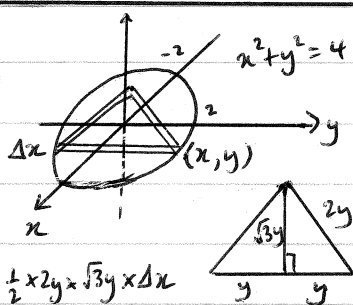
$= (\frac{1}{2}(p+q), \frac{p+q}{2pq})$

$= (\frac{1}{2}(p+q), 1)$

$\therefore$ Locus of M is $y = 1$

but only for $x < 0, x > 1$

(6)(a)



$$\Delta V = \frac{1}{2} \times 2y \times \sqrt{3}y \times \Delta x$$

$$= \sqrt{3}(4-x^2)\Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{-2}^2 \sqrt{3}(4-x^2)\Delta x$$

$$= 2\sqrt{3} \int_0^2 (4-x^2) dx$$

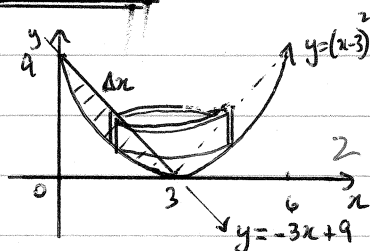
$$= 2\sqrt{3} \left[4x - \frac{1}{3}x^3 \right]_0^2$$

$$= 2\sqrt{3} \left(8 - \frac{8}{3} \right)$$

$$= \frac{32\sqrt{3}}{3} \text{ cubic units.}$$

4

(6)(b)



$$(ii) \Delta V = 2\pi(3-x)(-3x+9-(x-3)^2)\Delta x$$

$$= 2\pi(3-x)(3x-x^2)\Delta x$$

$$= 2\pi(9x-6x^2+x^3)\Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_0^3 2\pi(9x-6x^2+x^3)\Delta x$$

$$= 2\pi \int_0^3 (9x-6x^2+x^3) dx$$

$$= 2\pi \left[\frac{9}{2}x^2 - 2x^3 + \frac{1}{4}x^4 \right]_0^3$$

$$= \frac{27\pi}{2} \text{ cubic units.}$$

3

(c)(i) C is $\left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right)$

1

$$(ii) \Delta V = \pi[(1-\sin x)^2 - (1-\cos x)^2]\Delta x$$

$$= \pi[1-2\sin x + \sin^2 x - (1-2\cos x + \cos^2 x)]\Delta x$$

$$= \pi[2\cos x - 2\sin x - (\cos^2 x - \sin^2 x)]\Delta x$$

$$= \pi[2\cos x - 2\sin x - \cos 2x]\Delta x$$

2

$$(iii) V = \lim_{\Delta x \rightarrow 0} \sum_0^{\frac{\pi}{4}} \pi[2\cos x - 2\sin x - \cos 2x]\Delta x$$

$$= \pi \int_0^{\frac{\pi}{4}} (2\cos x - 2\sin x - \cos 2x) dx$$

$$= \pi \left[2\sin x + 2\cos x - \frac{1}{2}\sin 2x \right]_0^{\frac{\pi}{4}}$$

$$= \pi \left[\sqrt{2} + \sqrt{2} - \frac{1}{2} - (0+2-0) \right]$$

$$= \frac{\pi}{2} (4\sqrt{2}-5) \text{ cubic units.}$$

3

(7)(a)(i) Let $\alpha = \tan^{-1}(n+1)$, $\beta = \tan^{-1}n$

$$\Rightarrow \tan \alpha = n+1, \tan \beta = n$$

$$\text{Now } \tan(\alpha-\beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$= \frac{n+1-n}{1+(n+1)n}$$

$$\therefore \cot(\alpha-\beta) = 1+n+n^2$$

$$\therefore \alpha-\beta = \cot^{-1}(1+n+n^2)$$

$$\therefore \tan^{-1}(n+1) - \tan^{-1}n = \cot^{-1}(1+n+n^2)$$

$$(ii) \cot^{-1}(3) + \cot^{-1}(7) + \dots + \cot^{-1}(1+n+n^2)$$

$$= (\tan^{-1}2 - \tan^{-1}1) + (\tan^{-1}3 - \tan^{-1}2) + \dots$$

$$+ \dots + (\tan^{-1}(n+1) - \tan^{-1}n)$$

$$= \tan^{-1}(n+1) - \frac{\pi}{4}$$

2

$$(b)(i) \left(3 + \frac{1}{8}\right)^4 + \left(3 - \frac{1}{8}\right)^4$$

$$= 3^4 + 4 \cdot 3^2 \cdot 6 + \frac{4}{8^2} + \frac{1}{8^4} + 3^4 - 4 \cdot 3^2 \cdot \frac{1}{8} + \frac{1}{8^2} + \frac{1}{8^4}$$

$$= 2\left(3^4 + \frac{1}{8^4}\right) + 12$$

3

$$\therefore (2\cos \theta)^4 + (2\sin \theta)^4 = 4\cos 4\theta + 12$$

$$\therefore 16\cos^4 \theta + 16\sin^4 \theta = 4\cos 4\theta + 12$$

$$4\cos^4 \theta + 4\sin^4 \theta = \cos 4\theta + 3$$

$$(ii) 8x^4 + 8(1-x^2)^2 = 7$$

$$\text{Let } x = \cos \theta$$

$$8\cos^4 \theta + 8(1-\cos^2 \theta)^2 = 7$$

$$8\cos^4 \theta + 8\sin^4 \theta = 7$$

$$\Rightarrow 2(\cos 4\theta + 3) = 7$$

$$\Rightarrow \cos 4\theta = \frac{1}{2}$$

$$\therefore 4\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}, \dots$$

$$\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12}, \dots$$

$$\therefore \text{Solutions } x = \cos \frac{\pi}{12}, \cos \frac{5\pi}{12}, \cos \frac{7\pi}{12}, \cos \frac{11\pi}{12}$$

$$\text{ie } x = \pm \cos \frac{\pi}{12}, \pm \cos \frac{5\pi}{12}$$

3

$$(iii) 8x^4 + 8(1-x^2)^2 = 7$$

$$8x^4 + 8 - 16x^2 + 8x^4 = 7$$

$$16x^4 - 16x^2 + 1 = 0$$

$$16m^2 - 16m + 1 = 0 \quad (x^2 = m)$$

$$\text{has roots } \cos^2 \frac{\pi}{12}, \cos^2 \frac{5\pi}{12}$$

$$\therefore \cos^2 \frac{\pi}{12} \cos^2 \frac{5\pi}{12} = \frac{1}{16}$$

$$\therefore \cos \frac{\pi}{12} \cos \frac{5\pi}{12} = \frac{1}{4}$$

$$\cos \frac{\pi}{12} > 0$$

$$\cos \frac{5\pi}{12} > 0$$

Also

$$\left(\cos \frac{\pi}{12} + \cos \frac{5\pi}{12}\right)^2 = \cos^2 \frac{\pi}{12} + \cos^2 \frac{5\pi}{12} + 2\cos \frac{\pi}{12} \cos \frac{5\pi}{12}$$

$$= \frac{-(-16)}{16} + 2 \times \frac{1}{4}$$

$$= \frac{3}{2}$$

3

$$\therefore \cos \frac{\pi}{12} + \cos \frac{5\pi}{12} = \sqrt{\frac{3}{2}}$$

$$(iv) \cos \frac{\pi}{12} \text{ and } \cos \frac{5\pi}{12} \text{ are solutions}$$

to the equation:

$$x^2 - \sqrt{\frac{3}{2}}x + \frac{1}{4} = 0$$

$$\text{ie } 4x^2 - 2\sqrt{6}x + 1 = 0$$

$$x = \frac{2\sqrt{6} \pm \sqrt{24 - 4 \times 4 \times 1}}{2 \times 4}$$

$$= \frac{2\sqrt{6} \pm 2\sqrt{2}}{8}$$

$$= \frac{\sqrt{6} \pm \sqrt{2}}{4}$$

2

$$\therefore \cos \frac{\pi}{12} = \frac{\sqrt{6} + \sqrt{2}}{4}, \cos \frac{\pi}{12} > \cos \frac{5\pi}{12}$$

(8)(a) AUSTRALIA

No. of words = No with diff letters +

No with 2 A's +

No with 3 A's

$$= {}^7C_6 \times 6! + {}^6C_4 \times \frac{6!}{2!} + {}^6C_3 \times \frac{6!}{3!}$$

$$= 5040 + 5400 + 2400$$

$$= \underline{\underline{12840}} \quad 4$$

(b) $\frac{a+b+c}{3} > \sqrt[3]{abc}$, $a+b+c=1$
 $a>0, b>0, c>0$

(i) $\frac{a+b+c}{3} > \sqrt[3]{abc}$

$$\Rightarrow \frac{1}{3} > \sqrt[3]{abc}$$

$$\Rightarrow \frac{1}{27} > abc$$

$$\Rightarrow 27 < \frac{1}{abc}$$

$$\Rightarrow \frac{1}{abc} > 27 \quad 1$$

(ii) $\frac{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}{3} > \sqrt[3]{\frac{1}{abc}}$

$$\therefore \frac{1}{a} + \frac{1}{b} + \frac{1}{c} > 3 \times \sqrt[3]{\frac{1}{abc}}$$

$$> 3 \times \sqrt[3]{27}$$

$$> 3 \times 3$$

$$> \underline{\underline{9}} \quad 2$$

(iii) From (ii)

$$\frac{bc+ac+ab}{abc} > 9$$

$$\therefore ab+bc+ac > 9abc \quad (*)$$

Now $(1-a)(1-b)(1-c)$

$$= (1-a-b+ab)(1-c)$$

$$= 1-a-b+ab-c+ac+bc-abc$$

$$= 1-(a+b+c) + (ab+bc+ac) - abc$$

$$= (ab+bc+ac) - abc$$

$$> 9abc - abc \quad \text{from } (*)$$

$$> \underline{\underline{8abc}} \quad 2$$

(c) $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$

(i) $I_n = \int_0^{\frac{\pi}{2}} \sin^{n-1} x \cdot \sin x \, dx$

Let $u = \sin^{n-1} x$ $v' = \sin x$

$u' = (n-1) \sin^{n-2} x \cos x$ $v = -\cos x$

$$I_n = \left[-\sin^{n-1} x \cos x \right]_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cos^2 x \, dx$$

$$= 0 - 0 + (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x (1 - \sin^2 x) \, dx$$

$$= (n-1) [I_{n-2} - I_n]$$

$$\therefore [1 + (n-1)] I_n = (n-1) I_{n-2}$$

$$n I_n = (n-1) I_{n-2}$$

$$\therefore \underline{\underline{I_n = \left(\frac{n-1}{n} \right) I_{n-2}}} \quad 3$$

(ii) $\int_0^{\frac{\pi}{2}} \sin^{2n} x \, dx = I_{2n}$

$$= \frac{2n-1}{2n} \cdot I_{2n-2}$$

$$= \frac{(2n-1)(2n-3)}{2n(2n-2)} I_{2n-4}$$

$$= \frac{(2n-1)(2n-3)(2n-5) \dots 3 \times 1}{2n(2n-2)(2n-4) \dots 4 \times 2} I_0$$

$$= \frac{2n(2n-1)(2n-2)(2n-3) \dots 2 \times 1}{[2n(2n-2)(2n-4) \dots 4 \times 2]^2} I_0$$

$$= \frac{(2n)!}{(2^n)^2 [n(n-1)(n-2) \dots 2 \times 1]^2} I_0$$

$$= \frac{(2n)!}{2^{2n} (n!)^2} \int_0^{\frac{\pi}{2}} 1 \, dx$$

$$= \frac{(2n)!}{2^{2n} (n!)^2} \left[x \right]_0^{\frac{\pi}{2}}$$

$$= \frac{(2n)!}{2^{2n} (n!)^2} \times \frac{\pi}{2}$$

$$= \underline{\underline{\frac{\pi (2n)!}{2^{2n+1} (n!)^2}}} \quad 3$$